New Hope Bird Alliance - Habitat Trend Lines

Observed bird counts by location and season, 1980 onward

This document uses **only Joel's six habitat CSV files** in `habitat_data/`. Each graph shows observed annual counts (points) and a smooth descriptive trend for each habitat (lines). The curves are not adjusted for survey effort and are not estimates of true population size.

Save this `.qmd` file in the **main repository folder**, next to `new-hope-birds.Rproj`. You can run chunks individually using the green arrows, or click **Render** to generate the full report.

1. Load packages and set colors

```
library(tidyverse)

start_year <- 1980
habitat_colors <- c(
  "Forest birds" = "#009E73",
  "Grassland birds" = "#E69F00",
  "Aridland birds" = "#D55E00",
  "Wetland and coastal birds" = "#0072B2",
  "Tundra birds" = "#CC79A7",
  "Habitat generalists" = "#555555"
)

dir.create("habitat-trend-plots", showWarnings = FALSE)
```

2. Read Joel's data

Joel's CSVs have one row per habitat and one column per year. We turn those year columns into two variables, `year` and `count`, so ggplot can use them.

```r
read_habitat <- function(location, season) {
  file <- paste0("../habitat_data/", location, "_", season, "_habitat.csv")

  read.csv(file, check.names = FALSE) |>
    pivot_longer(cols = -habitat, names_to = "year", values_to = "count") |>
    mutate(
      year = as.integer(year),
      count = as.numeric(count),
      habitat = factor(habitat, levels = names(habitat_colors))
    ) |>
    filter(year >= start_year) |>
    arrange(habitat, year)
}
```

3. Make a simple trend-line graph

The dots show the observed counts. The smooth curves are LOESS trend lines, with no confidence interval shading. We use the same colors for all six graphs.

```r
plot_habitat <- function(location, season, title) {
  birds <- read_habitat(location, season)

  p <- ggplot(birds, aes(x = year, y = count, color = habitat)) +
    geom_point(alpha = 0.3, size = 0.8, na.rm = TRUE) +
    geom_smooth(method = "loess", formula = y ~ x,
                span = 0.75, se = FALSE, linewidth = 1, na.rm = TRUE) +
    scale_color_manual(values = habitat_colors, drop = FALSE) +
    scale_x_continuous(breaks = seq(1980, max(birds$year), by = 10)) +
    scale_y_continuous(labels = scales::comma) +
    labs(
      title = title,
      subtitle = paste("Habitat trends from 1980 to", max(birds$year)),
      x = "Year", y = "Observed bird count", color = "Habitat",
      caption = "Dots: annual counts | Lines: LOESS trends | Not adjusted for survey effort"
    ) +
    theme_minimal() +
    theme(legend.position = "bottom")

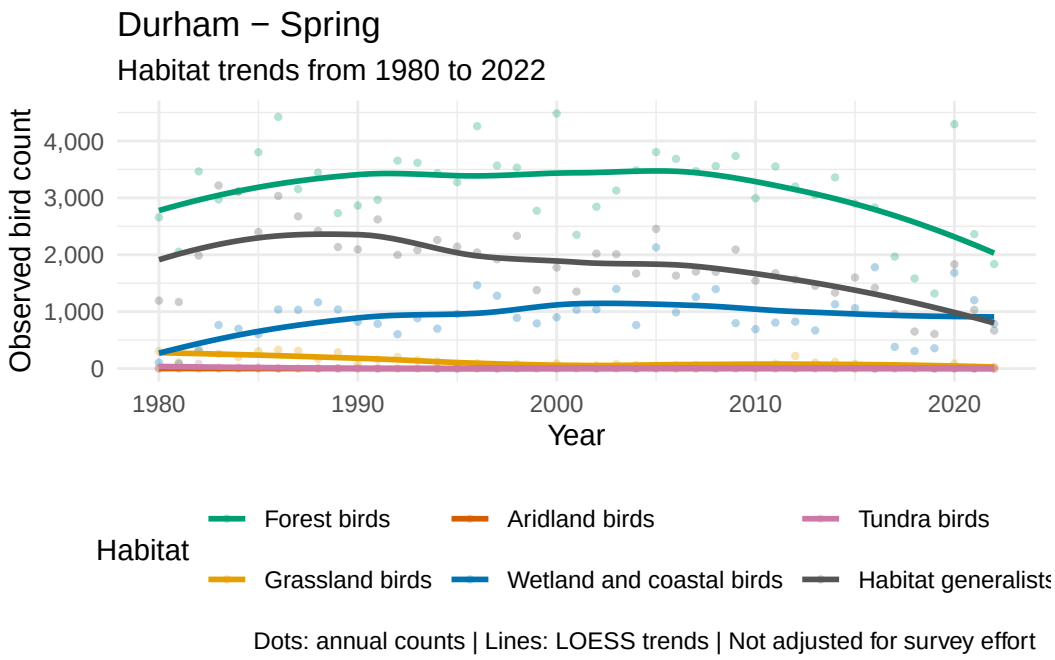
  ggsave(
    filename = paste0("habitat-trend-plots/", location, "_", season,
                      "_habitat_trendlines.png"),
```

```
    plot = p, width = 11, height = 7, dpi = 300
  )

  p  # Display the plot when the chunk runs
}
```

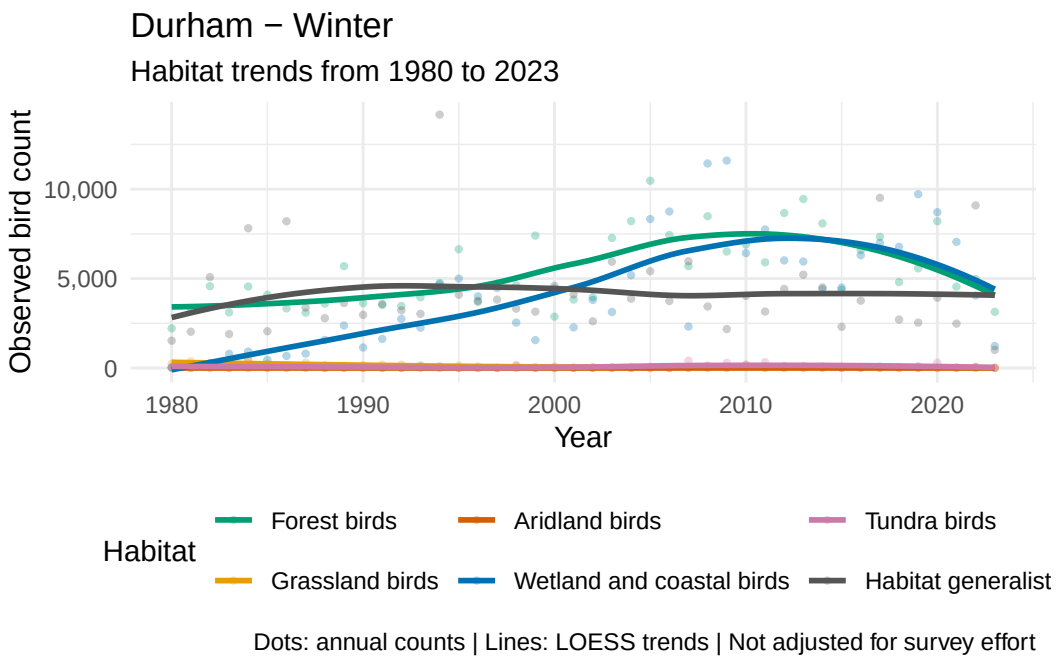
4. Durham — Spring

```
plot_habitat("durham", "spring", "Durham - Spring")
```



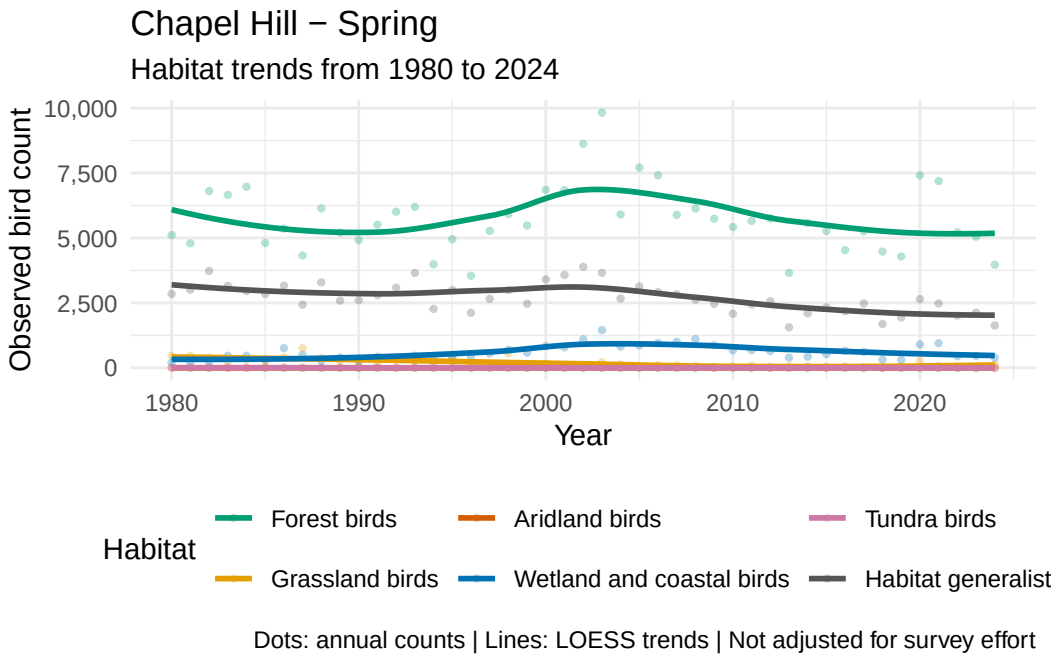
5. Durham — Winter

```
plot_habitat("durham", "winter", "Durham - Winter")
```



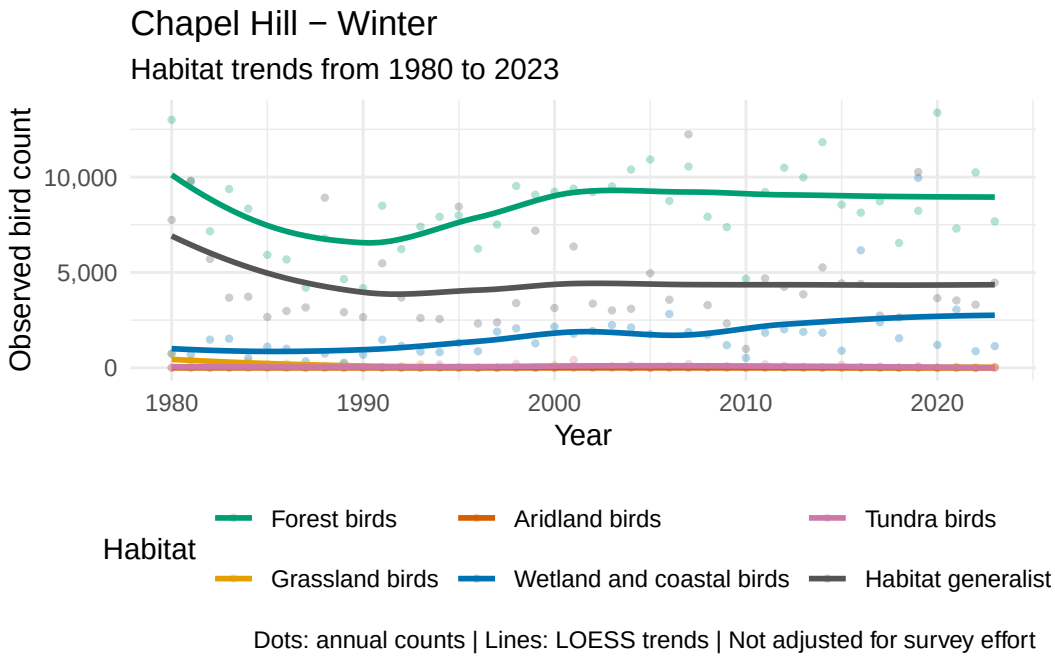
6. Chapel Hill — Spring

```
plot_habitat("chapel_hill", "spring", "Chapel Hill - Spring")
```



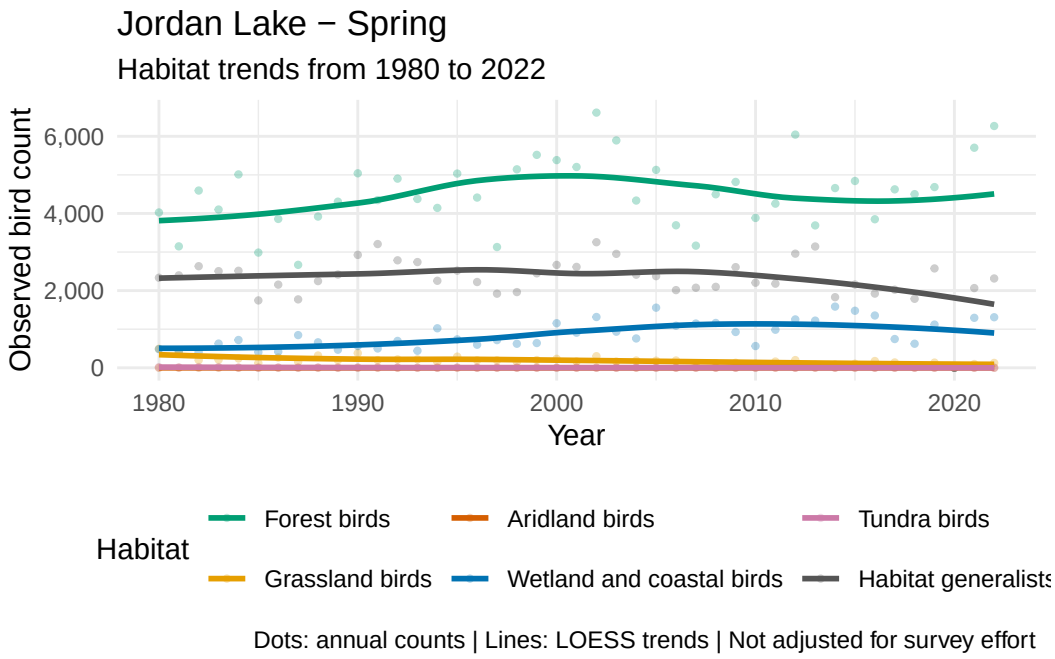
7. Chapel Hill — Winter

```
plot_habitat("chapel_hill", "winter", "Chapel Hill - Winter")
```



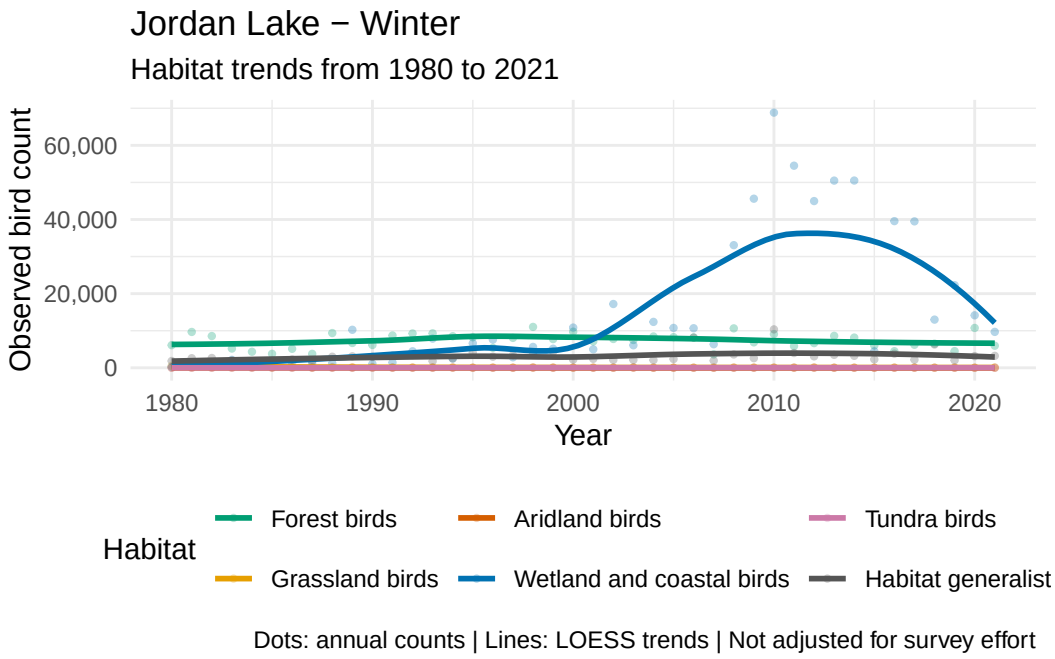
8. Jordan Lake — Spring

```
plot_habitat("jordan_lake", "spring", "Jordan Lake - Spring")
```



9. Jordan Lake — Winter

```
plot_habitat("jordan_lake", "winter", "Jordan Lake - Winter")
```



Notes and limitations

- These are **observed bird counts**, not survey-effort-adjusted population estimates.
- Some habitats have zero counts throughout, so their curves may overlap at zero.
- Jordan Lake spring has an all-zero year in 2020 in the supplied data; this should be checked with the team before interpreting it as a genuine decline.
- LOESS can produce fitted values below zero near very small counts. Such values are a smoothing artifact, not negative bird counts.
- Each location and season remains separate, as requested. No outside datasets are used.